

10 — Design System

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— consolidated from docs/design/README.md and v10-design/*.

1. Position

This section defines the **single source of design truth** for every HelixVPN user-facing surface: the Client (8 platforms), the Console (admin web), and the Connector (appliance UI).

It is a **spec-only** consolidation. The authoritative design files live under docs/design/ and the Volume-10 nano-detail docs; this README provides the navigable entry point and the architectural invariants.

2. Decoupled, reusable helix_design

The design system is a **standalone own-org submodule** at vasic-digital/helix_design, consumed as a flat submodule from the HelixVPN root. It is project-not-aware: HelixVPN brand values are injected as a swappable preset, never hardcoded inside the submodule.

Invariant	Why
Project-not-aware	Reusable across unrelated projects per §11.4.74
Flat layout	No nested own-org submodule chains per §11.4.28(C)
Equal codebase	Same test/doc/push discipline as the parent repo
Catalogue-first	Missing patterns are extended upstream, never forked in-project

The Flutter helix-ui workspace re-exports the generated Dart package; native shims consume the polyglot forms directly.

3. Token tiers

There is exactly one authored token source — JSON files in helix_design/tokens/**. Every consumable form is generated from it.

Tier	Role	Example
Primitive	Raw, theme-agnostic values	color.primitive.helix.500 = #3D5AF1
Semantic	Intent, light + dark fork	color.semantic.state.connected.fill
Component	Widget slot, references semantic only	comp.connectButton.bg.connected

Rules:

- Primitive tokens are literals; semantic tokens reference primitives; component tokens reference semantic tokens only.
 - Every semantic color/elevation token defines **both** light and dark values.
 - Hand-editing a generated form (CSS, Dart, etc.) is a drift defect.
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4. Export forms

The canonical JSON source is emitted into every target form so no platform re-types a color by hand.

Target	Output	Consumer
CSS	dist/css/helix.css	Console web, OpenDesign round-trip
Dart	dist/dart/helix_tokens.dart + ThemeExtension	Flutter helix-ui
SwiftUI	dist/swift/HelixTokens.swift	iOS/macOS NE config UI
Jetpack Compose	dist/compose/HelixTokens.kt	

Target	Output	Consumer
		Android quick-settings / notifications
ArkTS	dist/arkts/helix_tokens.ets	HarmonyOS native surfaces
C / Qt	dist/cqt/helix_tokens.h + .qml	Aurora OS

dist/** is a build-derivative; the source (tokens/**, presets/**, export/**) is tracked.

5. Component & screen index

The component library and screen wireframes are catalogued under docs/design/:

Area	Docs
Component specs	docs/design/components/README.md — desktop, mobile, Aurora, web
Screen wireframes	docs/design/screens/README.md — every screen across all platforms
Interaction patterns	docs/design/interaction/README.md — animation, gestures, accessibility
Assets & icons	docs/design/exports/ — PNG/PDF component and screen exports

Signature components include: ConnectButton, StatusChip, ExitPicker, ShieldIndicator, AdaptiveScaffold, and the connection-state palette mapped to the 7-variant FFI TunnelStatus.

6. OpenDesign integration

OpenDesign (submodules/open-design) is the mandatory design-and-refinement system per §11.4.162. helix_design authors an OpenDesign-native design system under helix_design/opendesign/helix/ (DESIGN.md + tokens.css + manifest.json + components.html).

- OpenDesign owns **authoring, refinement, and the CSS/markdown design-system form.**
- helix_design owns the **canonical JSON source and the polyglot distribution.**

- If OpenDesign lacks a needed pattern, the fix is a PR upstream to [nexu-io/open-design](#), recorded with a Catalogue-Check: `extend nexu-io/open-design@<sha> line`.

Current status:

- OpenDesign submodule built and daemon starts.
- HelixVPN design system imports successfully.
- Token contract validation reports needs-rebuild (low source-backed A1 coverage).
- No built-in Figma/Sketch/Adobe/XD/Penpot exporters; only Figma import is available.

For full validation results, see [docs/reviews/mvp-final/findings/phase2-opendesign-report.md](#).

7. Quality gates owned here

Gate	Asserts
DS-DRIFT	Generated forms are byte-identical to a fresh pnpm export
DS-LIGHT-DARK-COMPLETE	Every semantic color/elevation token defines both themes
DS-CONTRAST	Every on-X / X pair clears WCAG AA in light and dark
DS-NO-OVERLAP	Golden screenshots show no overlapping elements / labels
DS-VISUAL-REGRESSION	Per-component golden screenshots match baseline
DS-DECOUPLING	A non-HelixVPN preset builds green with zero HelixVPN refs in machinery
DS-EXPORT-FORM-VALID	Each emitted form parses in its target toolchain
DS-SCHEMA	tokens/** validate against JSON schema

8. Cross-references

- Design-system master index → [docs/design/README.md](#)
- Submodule overview → [../v10-design/00-overview-and-submodule.md](#)
- OpenDesign foundation → [../v10-design/opendesign-foundation.md](#)
- Design tokens → [../v10-design/design-tokens.md](#)

- Client core & UI → 05 — Client Core & UI

Sources: docs/design/README.md, v10-design/.md.*